
EDUCATION

Stanford University

Stanford, CA

B.S. **Mechanical Engineering**; Graduating in June 2027; GPA: **3.742**

Sep 2023 – Present

Relevant coursework: ME 30: Thermodynamics, ME 70: Fluid dynamics, ME 80: Mechanics of Materials, ME 149: Mechanical Measurements, ME 102: Foundations of Product Realization, ENGR 40 M, ENGR 15: Dynamics, AA126: Aerospace Prototyping Lab, ME161: Dynamics Systems, Vibrations and Controls

SKILLS

- **Programming & Controls:** Python, Arduino, MATLAB, JavaScript, C++
- **CAD & Simulation:** Fusion360, Onshape, SolidWorks, CNC, Luminary Cloud, ANSYS Fluent, HyperMesh, OptiStruct
- **Fabrication:** CNC machining, 3D printing, laser cutting, soldering, sand casting, welding
- **Project Portfolio:** <https://mari-zampa.github.io/html-css-js-portfolio-mari/>

WORK EXPERIENCES

Bulwark Dynamics

Menlo Park, CA

Mechanical Engineer Intern

Dec 2025 – Present

Bulwark Dynamics designs and builds autonomous vessels for logistics, transport, and defense applications

<https://www.bulwarkdynamics.com/>

- Integrating mechanical and electrical subsystems on a 15-ft autonomous vessel prototype
- Designed and fabricated lightweight mounts and brackets for sensors and electronic.
- Routed and secured wiring harnesses, improving reliability and serviceability in high-vibration environments.
- Supported rapid testing and iteration to fix fit and alignment issues.

Stanford Shape Lab

Stanford, CA

Undergraduate Research Assistant

Dec 2023 – Present

Stanford Mechanical Engineering Department laboratory focused on the development of cutting-edge robotics, mechatronics, and sensing technologies with the primary goal to create dynamic 3D displays and haptic interfaces

<https://shape.stanford.edu/>

- Designed and prototyped wearable haptic devices integrating motors, drivers, and control electronics
- Debugged power, signal noise, I²C conflicts, and vibration performance across iterations
- Set up the Arduino and Node.js backend that connects the wristband to the web app, letting it send real-time vibration feedback based on handwriting errors.

Stanford Solar Car Project

Stanford, CA

Member of Aero Sub team, Mechanical Division

Mar 2025 – Present

Student-run organization at Stanford University dedicated to designing and building solar-powered cars including work on the most recent entry in the 2025 Formula Sun Grand Prix in Kentucky (2nd place)

- Integrated aerobody design with chassis, roll cage, and suspension hardpoints
- Ran aerodynamic simulations evaluating drag, separation behaviour, and crosswind/yaw stability.
- Built the CFD workflow for meshing, solver setup, and analysis of pressure distribution and wake structure.

PROJECTS

Mechanical Design Project

Apr – Jun 2025

ME 102, Stanford University

- Built a lever-actuated escapement and multi-stage gear train (planetary, bevel, worm-wheel). Performed kinematic, stress, and tolerance analyses. Fabricated via 3D printing in PLA and laser cutting.

Mechanical Measurement Labs

Apr – Jun 2025

ME 149, Stanford University

- Conducted rheology studies of non-Newtonian fluids, tensile testing of polymers, heat transfer experiments (Biot numbers, convection coefficients), and PIV vortex-ring analysis.

Wing Test Project

Sep – Nov 2025

AA126, Stanford University

- Designed and fabricated an airfoil wing section with custom mounting hardware, then calibrated the load-cell system and ran wind-tunnel tests across multiple angles of attack to analyze lift and drag performance.

AA126: Wind Tunnel Project

Sep 2025 – present

AA126, Stanford University

- Constructing a wind-tunnel assembly by CNC-cutting and heat-forming acrylic panels to build the contraction, test-section, and diffuser geometry, integrating a honeycomb flow straightener to condition the inlet stream and ensure uniform velocity profiles.